

SUNEEL KUMAR BURI

DATA ENGINEER | CLOUD & AI DATA ENGINEER

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years of experience designing scalable ETL/ELT pipelines, cloud data platforms, lakehouse architectures, data solutions supporting analytics and AI use cases. Proficient in Python, SQL, PySpark, Azure Databricks, Azure Data Factory, AWS Glue, Snowflake, Apache Airflow, and Delta Lake. Experienced in building governed, production-ready datasets that support analytics, machine learning, predictive modeling, and retrieval-augmented generation (RAG) applications.

TECHNICAL SKILLS

Programming & Processing: Python, SQL, PySpark, Spark SQL, Apache Spark, Bash

Cloud & Data Platforms: Azure Data Factory, Azure Databricks, Azure Data Lake Storage, Delta Lake, Azure Synapse Analytics, AWS Glue, Amazon S3, Amazon Redshift, Amazon EMR, Snowflake

Data Engineering: ETL/ELT, Batch and Streaming Pipelines, Apache Kafka, Apache Airflow, dbt, Lakehouse and Medallion Architecture, Data Integration, Data Migration

Databases & Modeling: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Dimensional Modeling, Star Schema, Slowly Changing Dimensions

AI, Quality & DevOps: Feature Engineering, MLflow, RAG Data Pipelines, Embeddings, Vector Search, Data Validation, Data Lineage, Git, Azure DevOps, Docker, Terraform, CI/CD, RBAC, PII Protection

PROFESSIONAL EXPERIENCE

ANALOG DEVICES | July 2025 - Present

AI Data Engineer

- Design and maintain scalable batch and near-real-time data pipelines using Python, SQL, PySpark, Azure Data Factory, and Azure Databricks for manufacturing, product-testing, supply-chain, and operational data.
- Build Azure lakehouse solutions with Azure Data Lake Storage, Delta Lake, and medallion architecture, creating governed bronze, silver, and gold datasets for analytics and AI workloads.
- Optimize Spark transformations, partitioning, caching, joins, and query plans, reducing average processing time by approximately 32% across high-volume engineering pipelines.
- Develop reusable ingestion frameworks for 15+ structured and semi-structured sources, including relational databases, REST APIs, JSON, CSV, Parquet, and equipment-generated files.
- Engineer feature pipelines and curated datasets supporting predictive maintenance, anomaly detection, quality analysis, demand forecasting, and product-performance modeling.
- Build document-ingestion workflows for Generative AI and RAG use cases, including text extraction, cleansing, chunking, metadata enrichment, embedding generation, and vector indexing.
- Implement schema validation, duplicate detection, null checks, referential-integrity rules, and reconciliation controls, improving early identification of data-quality issues by approximately 25%.
- Establish Git-based development, automated testing, Azure DevOps CI/CD, environment-specific configuration, monitoring, and production-support practices for reliable data releases.
- Collaborate with data scientists, analysts, and application teams to translate analytics and machine-learning requirements into scalable data models and production-ready pipelines.

Mastek | June 2021 - July 2023

ETL Developer | Cloud Data Integration

- Developed and supported cloud-based ETL/ELT pipelines using Python, SQL, AWS Glue, Amazon S3, Amazon Redshift, and Apache Airflow for customer, transaction, financial, and operational data.
- Integrated data from Oracle, SQL Server, MySQL, REST APIs, flat files, and third-party applications into standardized datasets for analytics, reporting, and downstream systems.
- Built 12+ scheduled workflows with dependency management, retries, logging, alerts, parameterized execution, and recovery procedures using AWS Glue and Apache Airflow.
- Designed fact tables, dimension tables, star schemas, and Slowly Changing Dimension processes to support historical reporting and business-intelligence requirements.
- Improved SQL queries, joins, indexing, partitioning, and incremental-loading logic, reducing processing time for selected daily loads by approximately 28%.
- Used PySpark and Amazon EMR to cleanse, transform, aggregate, and validate datasets that exceeded traditional single-node processing capacity.
- Automated record-count validation, file reconciliation, exception reporting, and source-to-target comparisons with Python and SQL, reducing manual verification effort by approximately 20%.
- Investigated pipeline failures, schema changes, missing records, duplicate data, and performance issues; documented root causes and implemented preventive improvements.
- Translated source-to-target requirements into data mappings, transformation rules, technical specifications, and test scenarios in partnership with business analysts and QA teams.

SELECTED PERSONAL DATA ENGINEERING PROJECTS

REAL-TIME INDUSTRIAL EQUIPMENT ANOMALY-DETECTION PLATFORM | Python, Apache Kafka, PySpark, Databricks, Delta Lake, MLflow

- Designed a real-time data pipeline that ingests simulated equipment telemetry through Apache Kafka and processes events with Spark Structured Streaming.
- Implemented event-time processing, checkpointing, deduplication, watermarking, schema enforcement, and late-arriving data handling to improve pipeline resilience.
- Created bronze, silver, and gold Delta Lake tables for raw telemetry, validated sensor readings, engineered features, and anomaly-detection results.
- Developed feature-engineering workflows for temperature, vibration, pressure, and equipment-utilization data and tracked machine-learning experiments with MLflow.
- Added pipeline monitoring, error handling, data-quality checks, and restart capabilities to support reliable streaming-data operations.

ENTERPRISE DOCUMENT RAG DATA PIPELINE | Python, LangChain, Databricks, Azure Data Lake Storage, Vector Search

- Built an end-to-end ingestion pipeline for PDF, text, JSON, and knowledge-base documents used by a Retrieval-Augmented Generation application.
- Developed document parsing, text cleansing, semantic chunking, metadata enrichment, embedding generation, and vector-indexing workflows.
- Implemented incremental processing and document-version controls to update changed content without rebuilding the complete knowledge index.
- Added metadata filtering, source attribution, duplicate-content detection, and retrieval-quality validation to improve response relevance and traceability.

EDUCATION

MASTER OF SCIENCE IN INFORMATION SCIENCE | August 2023 - May 2025

Saint Louis University, Saint Louis, MO

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE | August 2018 - May 2022

Vishnu Institute of Technology, India